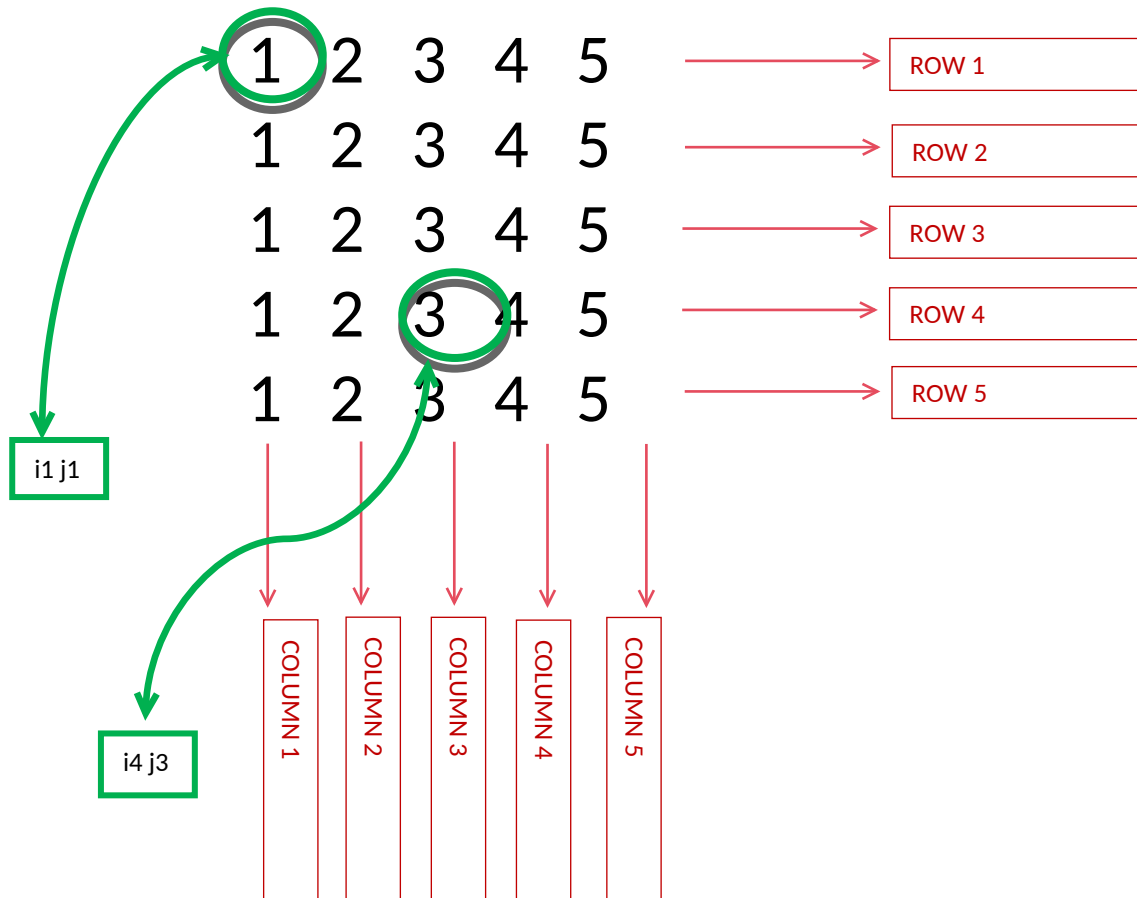


WEEK 01 MODULE
Programming Practice

TOPIC : 2 'Number Pattern'

6. NUMBER SQUARE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms a perfect square number pattern.
- number of rows is equal to the number of columns
- Every row contains the same sequence of numbers starting from 1 to n.
- There are no empty spaces, boundary checks, increasing sequences, decreasing sequences, or special conditions involved.

ii. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a two-dimensional grid (matrix)
- size = $n \times n$.
- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

For $n=5$

Each row is identical:

Row 1 → 1 2 3 4 5

Row 2 → 1 2 3 4 5

Row 3 → 1 2 3 4 5

Row 4 → 1 2 3 4 5

Row 5 → 1 2 3 4 5

Observations :-

- Number of rows = n
- Number of columns = n
- Each row prints full sequence from 1 to n
- Value depends only on column index j

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

- for ($i = 1$ to n)

Step 3: For every row generated by the outer loop, traverse all columns of that row using the inner loop.

-The inner loop is responsible for generating columns.

- for ($j = 1$ to n)

Step 4: Print value = j

- Every row prints 1 to n sequence

Step 5: After all numbers of the current row are printed, move to the next line.

Step 6: Repeat the same process until all n rows have been completed.

2. Program :-

```
package week1_module;
import java.util.Scanner;
public class NumberSquare {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of the square: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= n; j++) {
                System.out.print(j + " ");
            }

            System.out.println();
        }

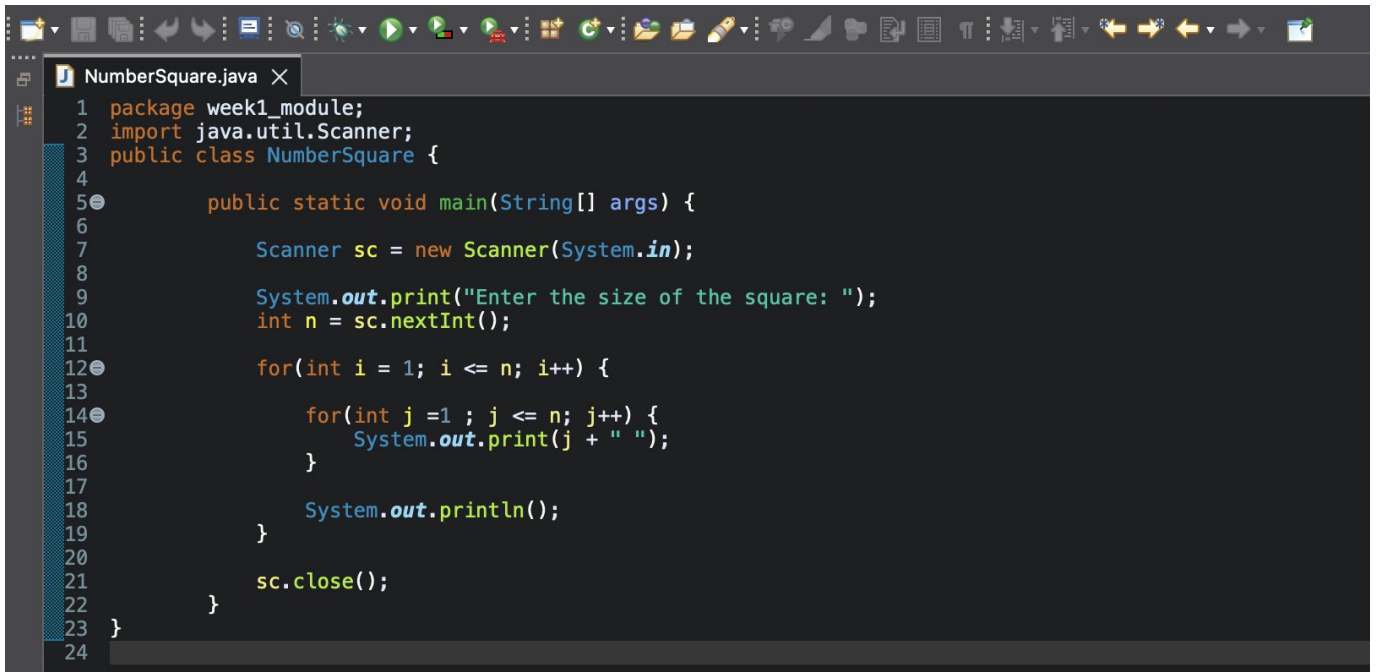
        sc.close();
    }
}
```

3. Program Output :-

```
Enter the size of the square: 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

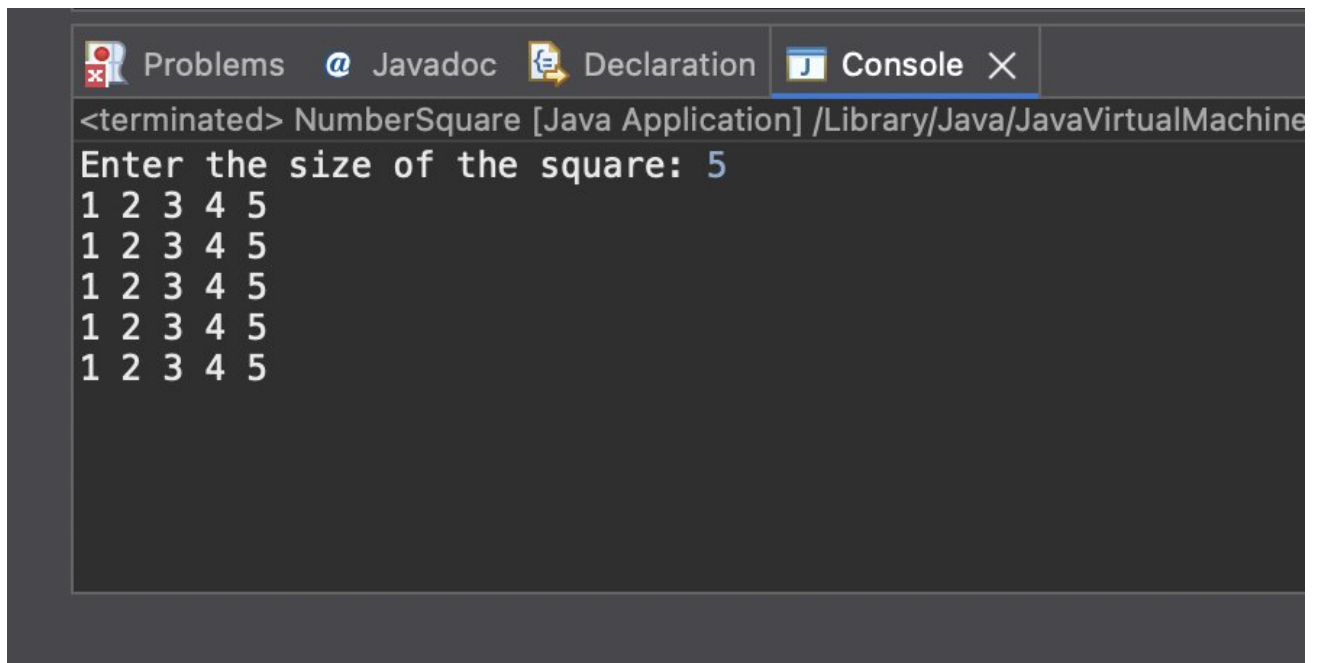
4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class NumberSquare {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the size of the square: ");
10        int n = sc.nextInt();
11
12        for(int i = 1; i <= n; i++) {
13
14            for(int j = 1; j <= n; j++) {
15                System.out.print(j + " ");
16            }
17
18            System.out.println();
19        }
20
21        sc.close();
22    }
23 }
24
```

ii. Program's Output :-



```
<terminated> NumberSquare [Java Application] /Library/Java/JavaVirtualMachine
Enter the size of the square: 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```